

Aadhithya Pattabiraman

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Summary

Data Analyst with 2+ years of experience designing and deploying machine learning models, natural language processing applications, and recommendation systems. Skilled in Python, SQL, and PySpark with hands-on expertise in feature engineering, model evaluation, and cloud-based deployment on AWS and Azure. Strong background in turning complex datasets into predictive insights and production-ready solutions, with a passion for applying advanced analytics to solve real-world business problems.

Education

New Jersey Institute of Technology

Master of Science in Data Science

May 2025

GPA: 3.9/4.0

Amrita Vishwa Vidyapeetham University

Master of Technology in Artificial Intelligence (AI)

May 2020

GPA: 3.6/4.0

Experience

Intuitive Surgical Inc.

Fremont, CA, USA

IT Engineering Intern

Sept 2024 – May 2025

- **License Usage:** Automated license usage reporting by integrating **OpenLM-API** with Python scripts, reducing manual reporting time by **60%** and ensuring daily data refresh. Improved data visibility and decision-making by creating and maintaining interactive **Tableau dashboards** for license utilization, enabling stakeholders to track trends across departments in real time.
- **Optimized data workflows** by building scheduled pipelines in **Databricks** and **Azure Data Factory** for automated ETL.
- **Cloud Storage and Analysis:** Ensured data schema consistency and performed data aggregation, trend analysis, and anomaly detection logic to spot unusual license activity using **PySpark** and **Pandas**. Enhanced cloud-based storage and access by integrating **Azure Blob Storage**, enabling secure, scalable data retention for analytics.
- **Optimization:** Streamlined **API-based** data retrieval by designing Python modules(*requests*, *schedule*) for license-usage queries, improving pipeline execution by **40%**. Identified **15%** underutilized licenses, contributing to budget optimization recommendations.

Nokia Solutions and Networks

Chennai, India

Software Developer - DevOps Team

Nov 2021 – Jan 2023

- **Web Framework Migration:** Leveraged the **Robot Framework** to design, develop, and execute test cases, demonstrating solid proficiency in test automation and a strong understanding of software testing concepts. Designed and executed **200+** automated test cases, cut test execution time by **80%** from **3 min** to **30s** using optimized **Selenium-Python** scripts.
- **Python Automation and Migration:** Created automated test result aggregation scripts in **Python**, parallel test execution, and test data management in Azure Repos, exporting execution metrics to **Power BI** for faster defect analysis and reporting.
- **Accelerated CI/CD** efficiency by **70%** by integrating automated test workflows into **Azure DevOps** pipelines, reducing manual QA effort. Improved release stability by detecting **40+** critical regression bugs and dependency-related failures before deployment.
- **Collaboration:** Enhanced cross-team collaboration by maintaining version-controlled test artifacts in Git and automating build-triggered test runs in **Azure** Pipelines, reducing deployment delays by **25%** and ensuring consistent test execution across releases.

Technical Skills

Programming: Python, R, Java, C++, Bash, PySpark, Django.

Database: MySQL, PostgreSQL, SQLite, MongoDB.

Data Services: Snowflake, AWS (Lambda, S3, EC2), Kafka, Airflow.

DevOps/Workflow: Git, Docker, CI/CD, REST API development.

Visualization/Packages: Tableau, Plotly, Power BI, Matplotlib, Seaborn, Scikit Learn.

ML/DL: PCA, Feature Engineering, CNN, RAG, NLP (BERT, SpaCy), Prompt Engineering.

Projects

Resume Parser & Job Matcher Using NLP

Jan 2024 – June 2024

- **Ranking:** Reduced candidate screening time by **60%** by automating resume ingestion, parsing, and semantic ranking using **LLM** powered document understanding, streamlining job fit predictions from **unstructured resumes**, to focus on high-fit candidates.
- **Algorithm:** Achieved **92%** entity extraction accuracy for skills, education, job titles and dates using **NLP** by combining **spaCy NER** with **pattern-based regex** rules and custom post-processing for noisy PDF text.
- Enhanced recruiter trust with explainable **AI**, **LLM-generated** fit summaries achieved **85%** human alignment in blind tests.
- Processed 10k+ documents and served 2s query responses via **MongoDB** indexing, **FastAPI**, and Dockerized deployment on AWS.

Cloud-Optimized Hybrid Movie Recommendation System

July 2021 – July 2022

- **General Approach:** Designed and implemented a hybrid recommender system using the MovieLens 100K dataset, combining **collaborative filtering** (user-user & item-item) and **content-based filtering** for personalized movie recommendations.
- Applied **Cosine Similarity**, **Pearson Correlation**, and **matrix factorization** via Surprise SVD to optimize user-item matching.
- **Cloud Integration:** Performed feature engineering on **100K+** records by integrating user demographics, genre embeddings, and movie metadata, increasing contextual recommendation accuracy. Achieved 200ms retrieving time for top recommendations using **AWS S3** for dataset storage, **AWS Lambda** for batch data preprocessing, and **AWS CloudWatch** for performance monitoring.
- Achieved **87%** precision, **84%** recall, and MAP score of **0.81** for **top-10** recommendations, outperforming baseline models by **22%**.
- **Enhancements:** Deployed the recommendation engine on **AWS SageMaker** with containerized inference endpoints to handle peak loads, scaling from 2 to 10 instances, achieving a 99.9% uptime and enabling real-time predictions with latency of **250ms**.